INCH-POUND

MIL-W-45991A(AL)
28 September 1989
SUPERSEDING
MIL-W-45991(AL)
25 June 1984

MILITARY SPECIFICATION

WELDING SHOP, TRAILER MOUNTED

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a Trailer Mounted Welding Shop, 350 Ampere Direct Current (DC), Constant Current/Constant Voltage (CC/CV) with all components necessary for Shielded Metal Arc Welding (SMAW), Gas Metal Arc Welding (GMAW), including shorting arc process, (DC) Gas Tungsten Arc Welding (GTAW), Flux Cored Arc Welding (FCAW), Air Carbon Arc Cutting and Gouging (AAC), Oxyacetylene Welding (OAW) and Oxyacetylene Cutting (OAC-A).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2.1).

SPECIFICATIONS

FEDERAL

BB-A-106	-	Acetylene, Technical, Dissolved
BB-C-101	-	Carbon Dioxide (CO_2): Technical and USP
BB-O-925	-	Oxygen, Technical, Gas and Liquid
TT-P-664	-	Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3431
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

PPP-B-1055 - Barrier Material, Waterproofed, Flexible
PPP-P-40 - Packaging and Packing of Hand Tools
PPP-T-60 - Tape: Packaging, Waterproof

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-T-152 - Treatment, Moisture-, and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173 - Varnish, Moisture- and Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Electrical Equipment)
MIL-B-208 - Battery, Storage, Lead Acid, Automotive and Navy Portable (Except Aircraft), Packaging and Packing of
MIL-C-3600 - Compressor, Rotary, Power Driven; and Compressors, Reciprocating, Power Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for
MIL-M-7866 - Molybdenum Disulfide Technical, Lubrication
MIL-E-10062 - Engines: Preparation for Shipment and Storage of
MIL-T-13880 - Torches, Oxyacetylene
MIL-A-18455 - Argon, Technical
MIL-W-45562 - Welding and Soldering Equipment, Supplies, and Accessories, Packaging of
MIL-W-45992 - Welding Machine, Arc, Direct Current (DC) Constant Current (CC), Constant Voltage (CV), 350 Ampere, Diesel Engine Driven (DED), Base Mounted
MIL-C-53072 - Chemical Agent Resistant Coatng (CARC) System Application Procedures & Quality Control Inspection

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Threads Standards For Federal Service

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of Military Property
MIL-STD-171 - Finishing of Metal and Wood Surfaces
MIL-STD-209 - Slinging & Tiedown Provisions for Lifting & Tying Down Military Equipment
MIL-STD-281 - Automobile, Trucks, Truck Tractors, Trailers and Trailer Dollies; Preservation and Packaging of
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-889 - Dissimilar Metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

and Marking
MS-53030 - Chassis, Trailer, Generator, 2-1/2 Ton, Military Design, M200A1

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government drawings and publications. The following other Government drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF ARMY

DRAWINGS

11021481 - Welding Machine Cover
11021251 - Finish Methods
11022000 - Welding Shop, Trailer Mounted
11022380 - Plate, Identification, Welding Shop
11022381 - Plate, Data, Welding Shop
11022560 - Gas Cylinder Cover
11022570 - Gas Cylinder Cover
13226E7140 - Pattern Camouflage Painting

LIST OF COMPONENTS

11021547 - List of Components

(Copies of specifications, standards, and other Government drawings and publications required by contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government Publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issue of the documents cited in the solicitation (see 6.2.1).

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

NEMA EW-1 - Electric Arc Welding

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 2101 L Street NW, Washington, DC 20037.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1160, 222 South Riverside Plaza, Chicago, IL 60606.)

AMERICAN WELDING SOCIETY (AWS)

AWS A3.0 - Terms and Definitions
AWS A5.18 - Specification for Carbon Steel Filler Metals for Gas Shielded Arc Welding

(Applications for copies should be addressed to the American Welding Society, 550 N.W. Le Jeune Road, PO Box 35140, Miami, FL 33135.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2.1),a sample shall be subject to first article inspection (see 6.3) in accordance with 4.2.1. The approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of the specification.

3.2 Design. The welding shop shall be in accordance with Army Drawing 11022000 and as specified herein. The shop shall include all tools and equipment specified in the list of components 11021547. Welding terms and definitions shall be in accordance with AWS A3.0 as referenced herein and on the applicable drawing.

3.3 Construction. The welding shop shall be free from any characteristics or defects which may render the shop unsuitable or inefficient for the purpose intended. The shop and its equipment shall be so constructed as to be capable of performing the operations specified herein.

3.3.1 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions specified herein. Materials shall be free from defects which would adversely affect the performance, reliability, durability, longevity, or maintainability of the individual components or overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1.1 Reclaimed materials. The manufacturer may use reclaimed materials for fabricating new parts. Reclaimed materials shall be reprocessed, remanufactured, or recycled in a manner that restores them to the same chemical composition and physical properties as the raw materials originally selected for use. Use of reclaimed parts as is or rebuilt from scrap or other used equipment shall not be permitted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

3.3.2 Castings and forgings. Finished castings and forgings shall be free of fins, scale, inclusions, cold shuts, voids, cracks, thermal ruptures, laps, folds, mismatching, or other defects. No processes such as welding, peening, plugging, soldering, or pasting shall be used for repairing, reworking, or improving cosmetic appearance.

3.3.3 Welding, brazing and soldering. All welding, brazing, and soldering shall be free of defects and of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. Defects shall include but are not limited to such things as bubbles cracks, gouges, holes, insufficient materials used. All flux or other such materials used during these operations shall be thoroughly removed (cleaned), from the part(s) immediately upon completion of the operation. These operations shall not be employed as repair measures for defective parts.

3.3.4 Fastening devices. Fastening devices and methods shall be chosen to serve the need while providing necessary adjustability for service, maintenance of repair. Fasteners shall be installed to prevent loss of tightness and shall not loosen in service.

3.3.5 Threads. All threaded parts of the welding shop shall conform to FED-STD-H28 and applicable "Detailed Standards" section referenced therein.

3.3.6 Cleaning, treating and painting. Surfaces to be painted shall be cleaned and dried to ensure they are free of oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion products of any other substances detrimental to the finish. Cleaning, pretreatment of surfaces, application of primer, and application of topcoat shall be in accordance with MIL-C-53072. Finish shall be as specified on the individual drawing, 11021251, and MIL-STD-171. A camouflage pattern shall be applied in accordance with drawing 13226E7140.

3.3.7 Fungus Resistance. Electrical and electronic components and electrical insulation not otherwise specified herein shall be fungus inert, in accordance with requirement 4 of MIL-STD-454. Components of electrical circuitry that are not weather resistant to moisture and fungus growth shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.3.8 Identification marking. The welding shop shall be marked for identification in accordance with MIL-STD-130 and shall conform with the applicable drawings and shall include the National Stock Number. The markings shall be applied to plates conforming to drawings 11022380 and 11022381.

3.3.9 Storage box locks. The equipment storage boxes shall each be furnished with a lock and two identical keys. The keys and lock shall be different for a series of no less than 30 units. The key series and lock shall repeat on each series of locks.

3.4 Performance and product characteristics.

3.4.1 Physical data.

3.4.1.1 Weight. The wet weight of the welding shop including all components and trailer chassis, shall not exceed 7410 pounds (3361 kilograms).

3.4.1.2 Lunette loading (wet weight). Weigh all components in place on the welding shop body and the welding shop body mounted on the trailer chassis, the vertical loading of the lunette eye shall not be greater than 635 pounds (288.03 kilograms) nor less than 350 pounds (158.76 kilograms).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

3.4.1.3 Dimensions. With the shop body mounted on the trailer chassis, the width shall not exceed 8 feet (2.438 meters), the length (including the trailer chassis) shall not exceed 14.9 feet (4.542 meters), and the height (including the trailer chassis) shall not exceed 8.6 feet (2.621 meters).

3.4.2 Transportability. Intended operation and function of welding shop components shall not be degraded as a result of transportation (such as cracked welds, permanent deformation, damaged equipment).

3.4.2.1 Lifting and tie-down. The weld shop as trailer mounted shall conform to the sling lifting and tie-down requirements of MIL-STD-209 for Type II equipment, Class 1 and Class 2. The trailer mounted weld shop shall be capable of external helicopter transport.

3.4.2.2 Roadability. The welding shop with trailer and fully loaded shall be capable of being towed at speeds up to 50 miles per hour (mph) (80.45 kilometers per hour (km/h)) on paved roads, up to 35 mph (56.33 km/h) on unpaved roads and up to 5 mph (8.05 km/h) on rough terrain.

3.4.2.3 Rail impact. The trailer mounted weld shop with all components properly stored for shipment shall withstand a rail impact test in accordance with MIL-STD-810, without damage or permanent deformation, at impact velocities of 4, 6, and 8 mph in forward direction, and 8 mph in reverse direction.

3.4.3 Environmental conditions.

3.4.3.1 Low temperature. The welding shop shall be capable of starting and operating down to a minimum temperature of -25 degrees Fahrenheit (F) (-31.67 degrees C). The welding shop shall not be adversely affected by storage down to a minimum temperature of -28 degrees F (-33.34 degrees C).

3.4.3.2 High temperature. The welding shop shall be capable of starting and operating up to a maximum temperature of +120 degrees F (+40.89 degrees C). The welding shop shall not be adversely affected by storage up to a maximum temperature of +160 degrees F (+71.11 degrees C), with the exception that gas cylinders shall be removed from the welding shop and stored at temperatures of no more than +120 degrees F (+40.89 degrees C). In addition, the welding shop shall not be adversely affected by a relative humidity between 80 and 100 percent during high temperature operation and storage.

3.4.4 Welding performance. The welding shop shall be capable of performing the following processes without malfunction when the power source is operated within the ranges as specified in NEMA-EW1 Class I (100) for the rating of the machine.

Shielded metal Arc Welding	- SMAW
Gas Metal Arc Welding, including shorting arc process	- GMAW
Flux Cored Arc Welding	- FCAW
(DC) Gas Tungsten Arc Welding	- GTAW
Air Carbon Arc Cutting and Gouging	- AAC

In addition, the welding shop shall be capable of performing the following processes in conformance with the requirements of MIL-T-13880, Type 1.

Oxyacetylene Welding	- OAW
Oxyacetylene Cutting	- OAC-A

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

3.5 Welding machine cover. Each welding shop shall be provided with a removable cover for the welding machine in accordance with Drawing Number 11021481.

3.6 Gas cylinder covers. Each welding shop shall be provided with removable covers for the gas cylinders in accordance with Drawing Number 11022560 for the right side gas cylinder (facing the welding machine control panel) and 11022570 for the left side gas cylinder.

3.7 Identical items. All welding shops of this classification furnished under any one contract shall be physically, functionally, and mechanically identical. This requirement includes parts, assemblies and components. Any deviation must have the prior written approval of the contracting officer.

3.8 Government furnished property. The shop shall be mounted on an M200A1 trailer chassis in accordance with MS-53030. Unless otherwise specified (see 6.2.1), one Chassis, Trailer, Generator, 2-1/2 Ton, Wheel, Military Design, M200A1 per shop set shall be furnished by the government.

3.9 Welding machine. Each welding shop shall include a welding machine conforming to MIL-W-45992.

3.10 Workmanship. The quality of workmanship imparted to the welding shop and all its component shall be equal to or exceed the quality of the current commercial welding shop manufactured by domestic producers of the type of welding shop addressed herein. Where the requirements herein exceed commercial requirements, the requirements herein shall take precedence. The neatness, safety, sharp edges, structural integrity (seams, laps, steps), material soundness (no cracks, deformations, structural weakness), brazes and soldering, painting, welding of the welding shop shall equal or exceed the workmanship requirements as stated herein. The quality of workmanship shall be such as to eliminate degradation of the form, fit, function, performance and appearance as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specieied in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to ensure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. When specified (see 6.2.1), first article inspection shall be performed on the welding shop, trailer mounted. This inspection shall include the examinations and tests specified in Table I.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to production units being offered for acceptance under the contract. The sampling for test and examination shall be in accordance with Table I. The contractor shall provide to the Government representative objective that the examination requirements have been met. Failure of a welding shop to meet the sample inspection tests shall be cause for the Government to suspend acceptance until:

a. The contractor has completed a failure analysis.

b. The applicable production processes and procedures to prevent recurrence of the same defect in future production are evaluated and corrected.

c. The welding shops awaiting submittal for Government acceptance, (including components and subassemblies at in-process or final assembly) are examined to ensure that the material containing the same defect(s) is purged from the inventory and not presented to the Government for acceptance.

d. Objective evidence of the results of the failure analysis and corrective actions are provided to the Government representative prior to submitting the reconditioned welding machine for retest.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

Table I. Tests and examination.

FIRST ARTICLE	QUALITY CONFORMANCE INSPECTION		TEST AND EXAMINATION	TEST PARA.	REQUIREMENT PARA.
	SAMPLING (1 OUT OF EA. 40)	100 PERCENT INSPECTION (EA. UNIT SHALL BE INSPECTED)			
X	-	-	Design (Drawing 10022000	4.3	3.2
X	-	X	Tools and Equipment	4.3	3.2
X	-	X	Construction	4.4	3.3
X	-	X	Materials	4.4	3.3.1
X	-	X	Casting and Forgings	4.4	3.3.2
X	-	X	Welding, Brazing and Soldering	4.4	3.3.3
X	-	X	Fastening Devices	4.4	3.3.4
-	-	X	Threads	4.4	3.3.5
X	-	X	Cleaning, Treating and Painting	4.4	3.3.6
X	-	X	Fungus and Moisture Resistance	4.4	3.3.7
X	-	X	Identification Marking	4.4	3.3.8
X	-	X	Storage Box Locks	4.4	3.3.9
X	-	-	Weight	4.5.1	3.4.1.1
X	-	-	Lunette Loading (wet weight)	4.5.2	3.4.1.2
X	X	-	Dimensions	4.5.3	3.4.1.3
X	-	-	Rail impact	4.6.1	3.4.2.3
X	-	-	Roadability	4.6.2	3.4.2.2
X	-	-	Lifting and tie-down	4.6.4	3.4.2.1
X	-	-	Low Temperature	4.7.1	3.4.3.1
X	-	-	High Temperature	4.7.2	3.4.3.2
X	X	-	Welding Performance	4.8	3.4.4
X	-	X	Welding Machine Cover	4.9	3.5
X	-	X	Gas Cylinder Cover	4.10	3.6
X	-	X	Identical Items	4.11	3.7
X	-	-	Welding Machine	4.12	3.9
X	-	X	Workmanship	4.13	3.10

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

4.3 Design. Visual examination shall be performed to ascertain compliance with Drawing 11022000 (see 3.2).

4.4 Construction. The inspection of the welding shop shall include visual and manual examination to determine conformance with the requirements of 3.3 through 3.3.9. Visual examination shall include verification of proper cleaning; presence of specified protective coatings; completeness of manufacture and assembly; adequacy of marking; and repetition of padlock key series. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at any practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the welding shop and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.5 Physical data.

4.5.1 Weight. The complete welding shop shall be weighed as per 3.4.1.1 using a scale. Weight shall be as specified in 3.4.1.1.

4.5.2 Lunette loading (wet weight). Using a scale, the lunette loading shall be as specified in 3.4.1.2.

4.5.3 Dimensions. Using a steel tape, the dimensions shall be as specified in 3.4.1.3.

4.6 Transportability. Tests shall be conducted in the following order: 4.6.1 followed by 4.6.2.

4.6.1 Rail impact. The welding shop (see 3.4.2.3) shall be subjected to a rail impact test in accordance with MIL-STD-810. The unpackaged welding shop shall be positioned lengthwise, centered on the longitudinal centerline of a railroad car. The railroad car carrying the welding shop shall be impacted once each into the buffer car at speeds of 4, 6, and 8 mph with the welding shop oriented in the forward direction and once at 8 mph with the welding shop oriented in the reverse direction. The rail impact procedure to be followed shall be in accordance with MIL-STD-810, Method 516.3, Procedure VIII. After the test, the welding shop shall be inspected visually for damage.

4.6.2 Roadability tests. To determine compliance with 3.4.2.2, the welding shop with trailer and fully loaded shall be towed:

a. On a paved road at speeds up to 50 mph (80.45 km/h) for a distance of 5 miles (8.06 kilometers).

b. On an unpaved road at speeds up to 35 mph (56.33 km/h) for a distance of 3 miles (4.83 kilometers).

c. On rough terrain at speeds up to 5 mph (8.05 km/h) for a distance of 3 miles (4.83 kilometers).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

After each of the above tests, the welding shop, all its components, and trailer shall be visually examined for any deleterious effects. In addition, after the rough terrain test, the welding machine (MIL-W-45992) shall be operated and visually inspected to determine that it produces welding power, auxiliary power and compressed air and that all instruments and gauges function properly. The welding performance tests (see 4.8) shall be performed upon completion of this test.

4.6.3 Lifting and tie-down test. To determine compliance with 3.4.2.1, the fully loaded welding shop shall be subjected to the following tests.

4.6.3.1 Slinging provisions. Each lifting provision, including the connection to the structural members, shall be tested. A static pull to no less than the required design load shall be conducted on each provision for no less than 90 seconds in a direction of pull 45 degrees upward from the welding shop floor toward the apex where the four lifting cables meet in flight. All provisions do not have to be tested at the same time. Tests shall be performed in accordance with MIL-STD-209. The design loads for the four lifting provisions are as follows:

a. Front two provisions - 9334 pounds
b. Rear two provisions - 6176 pounds

No permanent deformation or set or structural damage in the provisions or other equipment structural components shall result from applications of the loads to the provisions.

4.6.3.2 Tie-down provisions. Each tie-down provision, including the connection to the structural members, shall be tested. A static pull to no less than the required design load shall be conducted on each provision for no less than 6 seconds in the direction of pull indicated. All provisions do not have to be tested at the same time. Tests shall be performed in accordance with MIL-STD-209. The required design loads and directions of force for the six tie-down provisions are as follows:

a. Rear two provisions - 14,710 pounds - rear longitudinal (horizontal toward the rear, parallel to edge of bed)

b. Middle two and front two provisions - 7355 pounds - forward longitudinal

c. All six provisions - 2452 pounds - vertically downward

d. All six provisions - 3677 pounds - lateral (horizontally outward, perpendicular to longitudinal force in a)

No permanent deformation or set structural damage in the provisions or other equipment structural components shall result from application of the loads to the provisions.

4.7 Environmental conditions. Maximum power testing shall be performed for 5 continuous minutes at the low and high temperatures.

4.7.1 Low temperature. To assure compliance with 3.4.3.1, the complete welding shop shall be submitted to storage at a temperature of -28 degrees F (-33.34 degrees C) for a period of 24 hours. At the end of this period, the temperature shall rise to -25 degrees F (-31 degrees C) and after stabilization, the welding shop shall be started and allowed to warm up for at least 5 minutes, and the output of both the auxiliary power generator and the welding generator shall be checked for minimum and maximum output per NEMA EW-1 Class I (100), using a resistance load bank having a power factor of 0.99 or higher.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

4.7.2 High temperature. To assure compliance with 3.4.3.2, the complete welding shop (less gas cylinders) shall be submitted to storage at a temperature of +160 degrees F (+71.11 degrees C) and induced relative humidity of 80 to 100 percent for a period of 24 hours. At the end of this period, the temperature shall be reduced to +120 degrees F (+48.89 degrees C) and 80 to 100 percent relative humidity. After a 24 hour stabilization period, the welding shop shall be started and allowed to warm up at least 5 minutes, and the output of both the auxiliary power generator and the welding generator shall be checked for minimum and maximum output per NEMA EW-1 Class I (100) using a resistance load bank having a power factor of 0.99 or higher.

CAUTION: Remove all gas cylinders from the welding shop prior to the high temperature storage tests. The Compressed Gas Association publication G-1 limits storage of gases to a maximum of +120 degrees F (+49.89 degrees C).

4.8 Welding performance. To assure compliance with 3.4.4, the following tests shall be performed using the associated accessory items cited in the List of components 11021547, as required. All welding operations shall be performed by a trained operator. Welds on test plates shall be visually examined for acceptable quality. Failure of the welding shop to perform any of the following cited tests shall be cause for rejection (see 4.2.2).

4.8.1 Welding tests in the CC mode. Welding tests in the CC mode shall be as follows:

a. Using the cables specified in the welding shop (50 ft. (15.24 Meters) lengths), welding operations shall be performed using the SMAW process with 5/32 inch (3.97 millimeters (mm) diameter E-6010 electrodes on mild steel plates, 2 pieces approximately 2 inches (50.8 mm) by 10 inches (254 mm) by 1/4 inch (6.35 mm) in a horizontal fillet position, tee joint. Welds shall be 10 inches (254 mm) long.

b. Using the above cables, welding operations shall be performed using the SMAW process with 7/32 inch (5.56 mm) diameter E-7018 electrodes on mild steel plates, 2 pieces approximately 4 inches (101.6 mm) by 1-1/2 inches (38.1 mm) by 3/8 inches (9.53 mm) in a horizontal fillet position, tee joint. Welds shall be 4 inches (101.6 mm).

c. Using the cables, hose and air compressor specified in the welding shop, cutting and gouging shall be performed (AAC process) using 5/32 inch (3.97 mm) diameter copper coated carbon electrode and test plates of mild steel (2 pieces) approximately 4 inches (101.6 mm) by 10 inches (254 mm) by 1/2 inch (12.7 mm), cutting one piece across the 4 inch (101.6 mm) width and gouging the other 10 inch (254 mm) length.

d. Using the gas shielding tungsten electrode welding set (TIG torch), hose, cables and regulator specified in the welding shop, with Argon gas, 1/16 inch (1.59 mm) diameter 2 percent thoriated tungsten, perform a horizontal fillet lap joint weld using 2 pieces of 2 inch (50.8 mm) by 10 inch (254 mm) by 16 gauge stainless steel. Using 1/8 inch (3.18 mm) diameter 2 percent thoriated tungsten, perform a horizontal fillet joint weld using 2 pieces of 2 inch (50.8 mm) by 1/8 inch (3.18mm) stainless steel. Welds shall be 10 inches (254mm) long.

4.8.2 Welding tests in the CV mode. Welding tests in the CV mode shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

a. Using the welding set, gas shielded, consumable electrode, the welding cables, gas hose and the regulator/flowmeter specified in the welding shop, connected to a cylinder of carbon dioxide gas, perform a horizontal fillet tee joint weld using .035 inch (.89 mm) diameter per AWS A5.18 classification ER-70S-3 wire on test plates of 2 pieces of 2 inch (50.8 mm) by 10 inch (254 mm) by 1/8 inch (3.18 mm) mild steel. Welds shall be 10 inches (254 mm) long.

b. Using the welding set and accessories specified above, perform a horizontal fillet tee joint weld using 1/16 inch (1.59 mm) diameter, flux cored wire (Hobart 802 or equal) on test plates of 2 pieces of 4 inch (101.6 mm) by 10 inch (254 mm) by 3/8 inch (9.53 mm) mild steel. Welds shall be 10 inches (254 mm) long.

c. Using the welding set and accessories specified above, connected to a cylinder of Argon gas, perform a horizontal fillet lap joint weld using .035 inch diameter (.89 mm) aluminum welding wire ER-4043 on test plates consisting of 2 pieces of 2 inch (50.8 mm) by 10 inch (254 mm) by 1/8 inch (3.18 mm), 1100 series aluminum.

d. Using the welding set and accessories specified above, connected to a cylinder of Argon gas, perform a fillet lap joint weld using 3/64 inch (1.19 mm) diameter aluminum welding wire ER-4043 on test plates consisting of 2 pieces of 2 inch (50.8 mm) by 10 inch (254 mm) by 1/4 inch (6.35 mm), 1100 series aluminum. Welds shall be 10 inches (254 mm) long.

4.8.3 Oxyacetylene tests. A certificate of conformance shall be issued to ascertain compliance with 3.4.4 and the quality assurance provisions of MIL-T-13880.

4.9 Welding machine cover. Visual inspection for compliance with 3.5 shall be performed with a certificate of conformance issued to ascertain that the cover meets the requirements of 3.5 and component drawing requirements.

4.10 Gas cylinder covers. Visual inspection for compliance with 3.6 shall be performed with a certification of conformance issued to ascertain that the covers meet the requirements of 3.6 and component drawing requirements.

4.11 Identical items. A certificate of conformance shall be issued to ascertain compliance with paragraph 3.7.

4.12 Welding machine. Visual inspection for compliance with 3.9 shall be performed.

4.13 Workmanship. Quality of workmanship shall be visually inspected for compliance with 3.10.

4.14 Packaging inspection. The trailer mounted welding shop shall be inspected before and after packaging to determine compliance with the preservation, packing and marking requirements specified in section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2.1).

5.1.1 Level A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

5.1.1.1 Cleaning and drying. Cleaning and drying of surfaces of accessories and components, including assemblies that comprise the complete weld shop, shall be in accordance with the level A requirements and procedures of MIL-P-116. The cleaning and drying processes used shall in no way be injurious to the item or finish.

5.1.1.2 Preservation application. Preservatives and methods and submethods specified herein shall be in accordance with the requirements of MIL-P-116. The preservatives and method designated by symbol shall be in accordance with the corresponding symbol of MIL-P-116.

5.1.1.3 Unit protection. Wrapping material, cushioning, barrier material and unit containers specified herein shall be in accordance with the requirements in MIL-P-116. The arrangement of contents, blocking, bracing, cushioning, anchoring and waterproofing shall be in accordance with the criteria and guidelines of MIL-STD-1186 and MIL-P-116. Closure of the unit containers shall be in accordance with the appendix of the applicable container specification.

5.1.1.4 Sets, kits and outfits. Items and equipment in sets, kits and outfits that are furnished with a carrying case shall be individually coated with a preservative wrapped and replaced in the case. Carrying case shall be placed in appropriate storage area, as specified. The items shall be cushioned and secured in their cases and the cases secured to restrict movement and prevent damage inside the trailer.

5.1.1.5 Mounting brackets and storage provisions. When mounting brackets, devices, storage racks or storage compartments are furnished for the welding shop, items and equipment received from the supplier that have been preserved and provided unit protection shall be secured in the holding or storage facilities in a manner to control or restrict movement and prevent damage of the item during transport.

5.1.1.6 Covers. Each cover shall be folded in a manner to form a neat, compact unit pack, then wrapped in a barrier material conforming to PPP-B-1055, class E-2. The wrap shall be secured with tape conforming to PPP-T-60, type III. The item(s) shall be placed in a storage compartment or some other place provided and secured in such a manner to restrict movement and to prevent damage during transport.

5.1.1.7 Padlocks and keys. The interior of padlocks shall be coated with molybdenum disulfide, see MIL-M-7866, and the locks operated to assure coverage of all interior surfaces. The padlocks used for securing the lids shall be protected against the elements by wrapping the padlocks in barrier material conforming to MIL-B-121, Grade A. The wrap shall be secured with tape conforming to PPP-T-60. When all padlock keys have been identified, place them in the bag with the welder ignition key. The bag shall then be secured (taped, wired) inside one of the doors of the welder located toward the front of the trailer. The welder shall then be banded girthwise to secure the doors from being opened. Edge protectors and fiberboard pads shall also be used to prevent marring of the surface of the welder.

5.1.1.8 Compressor. Compressors and components for compressors shall be preserved and unit protection provided in accordance with level A requirements of MIL-C-3600

5.1.1.9 Welding machine. The welding machine and all component parts shall be preserved and provided unit protection in accordance with the level A requirements of MIL-W-45562.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

5.1.1.10 Drive Belts and pulleys. All drive belts shall be released from tension or removed from the assembly. Unprotected exterior bare ferrous metal surfaces of the pulley groove shall be coated with a thin film of rust inhibiting primer conforming to TT-P-664. Belts removed from an assembly shall be identified and processed method IC-1 and placed in one of the metal storage boxes provided.

5.1.1.11 Gas. The cylinders of argon, acetylene, carbon dioxide and oxygen shall be packaged, packed and marked in accordance with level A requirements of the applicable specification as follows:

Gas	Specification
Argon	MIL-A-18455
Acetylene	BB-A-106
Carbon Dioxide	BB-C-101
Oxygen	BB-O-925

5.1.1.12 Fire extinguisher. Any unpainted ferrous metal surfaces of the fire extinguisher requiring the application of a contact preservative shall be coated with type P-1. The fire extinguisher shall be mounted in the holding brackets on the inside cover of the front box and secured with tape conforming to PPP-T-60, type IV in a manner to restrict movement and prevent damage.

5.1.1.13 Hand tools and tool sets. Handtools such as clamps, pliers, gauges, electrical sanders, and tool sets shall be preserved and provided unit protection in accordance with level A requirements of PPP-P-40. The hand tools and tool sets, processed as specified above, shall be placed in the metal storage boxes provided.

5.1.1.14. Cable assemblies. Electric cable assemblies shall be processed Method III of MIL-P-116. External ends of the assemblies shall be wrapped with barrier material conforming to MIL-B-121, grade A, not less than 4 inches from each end of the cable and secured with tape conforming to PPP-T-60, type IV. The tape shall encircle the item girthwise and be positioned at the end of the barrier material in a manner to form a water tight wrap. Each cable assembly shall be coiled to a minimum safe diameter with no deforming or kinking and the coils secured with not less than four evenly spaced wraps of tape conforming to PPP-T-60, type IV. The coiled cable assemblies shall be stored in one of the metal storage boxes provided or mounted and secured in area provided, as applicable.

5.1.1.15 Hose and hose assemblies. Hose, hose assemblies and tubing, with or without fittings disconnected shall be provided unit protection in the same manner as the cable assemblies. The coiled hose assemblies shall be stored in one of the metal storage boxes provided or mounted and secured in area provided, as applicable.

5.1.1.16 Electrical wiring system. Exposed ends of wires, plug opening sockets, coupling plugs, terminals and openings in switches and junction boxes shall be covered and sealed with tape conforming to MIL-T-60, type IV. Any electric motors, blowers, relays or other electrical items shall have openings sealed to prevent foreign materials from entering. Items not attached to another assembly shall be stored in one of the metal storage boxes provided.

5.1.1.17 Openings. Large openings into the interior of the welding machine, compressor or other components of the weld shop shall be covered with barrier material conforming to PPP-B-1055, Class E-2 and secured in place with tape conforming to PPP-T-60, type IV. Small openings, except drain valves and drain cocks, shall be sealed with tape specified or closed with caps or plugs conforming to MIL-C-5501.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

5.1.1.18 Welding sets. All welding sets shall be processed in accordance with level A requirements of the welding PACK specification MIL-W-45562.

5.1.1.19 Miscellaneous welding supplies and equipment. Items, equipment and components not specifically covered by the preservation requirements of this document, shall be preserved as specified for supplies and equipment of a similar design and construction in accordance with the requirements of MIL-P-116 and MIL-W-45562.

5.1.1.20 Welding shop shrouded components. The welding machine, air compressor, tank, wire dispenser, and vise that have been anchored to prevent movement during transit, shall be individually covered with a waterproof wrap of barrier material conforming to PPP-B-1055, Class E-2. All other tools and equipment that are provided with another means of waterproofness or do not require waterproofness because of inherent reasons, shall be anchored to prevent movement during transit. Application and securing of the shrouds or waterproof wraps shall be in accordance with the provisions set forth in MIL-STD-1186. The shrouds and wraps shall be applied and secured in a manner so that the layout of the trailer components remains unchanged for easy access and does not interfere with maintenance operations required for upkeep of the components.

5.1.1.21 Consolidated metal storage boxes. The parts, components, accessories and assemblies preserved and provided unit protection as specified herein, except for those items mounted on the trailer (welding machine, wire dispenser, air compressor, vise and gas cylinders), shall be consolidated and placed in the waterproof metal storage boxes provided on the welding shop. Arrangement of contents, cushioning, blocking, bracing, anchoring provisions, and material requirements for inside the metal storage boxes shall be in accordance with MIL-STD-1186.

5.1.2 Level C. Level C preservation of the weld shop shall be in accordance with the requirements of MIL-STD-1190. This shall include the application of preservatives, unit protection, including blocking, bracing, cushioning, waterproofing. This shall also include banding of welding machine doors as specified in 5.1.1.7.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2.1).

5.2.1 Level A. The weld shop with equipment enclosed, mounted, secured and preserved, shall be prepared for unboxed or uncrated mobile shipment in accordance with MIL-STD-281.

5.2.2 Level B. Level B shall be prepared the same as for level A.

5.2.3 Level C. Level C shall be prepared for unboxed or uncrated mobile shipment shrouded in accordance with MIL-STD-1190 in a manner to assure safe delivery to destination and carrier acceptance in compliance with Uniform Freight Classification or National Motor Freight Classification rules.

5.3 Marking. The marking of interior and exterior containers shall be in accordance with the requirements of MIL-STD-129 and any special markings as required.

6. NOTES

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

6.1 Intended use. The equipment covered by this specification is intended to be used as a Trailer Mounted Welding Shop.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of the individual documents referenced (see 2.1.1 and 2.2).

c. First article, when required (see 3.1 and 4.2.1).

d. Government furnished property, if required (see 3.8).

e. First article inspection, when required (see 4.2.1).

f. Level of preservation and packing required (see 5.1 and 5.2).

6.2.2 Contract data requirements. Required technical data, such as operators manuals, parts lists, wiring diagrams, and other instructions for operation and maintenance as identified on a numbered DD Form 1664 should be specified on a DD Form 1423 incorporated into the contract.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provision of the Federal Acquisition Regulation. The first article should be a preproduction sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, test and approval of the first article.

6.4 Subject term (key word) listing.

Direct Current
Constant Current
Transportability
Environment
Generator
Compressor

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AL

Preparing Activity:
Army - AL

Project Number 3431-A198

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

STANDARDIZATION ACCOMPLISHMENT REPORT

Revision "A" to Military Specification MIL-W-45991A(AL),

Welding Shop, Trailer Mounted,

Project Number 3431-A198

The following changes have been effected by this revision:

a. The following reference documents have been deleted from this revision: MIL-P-10524/9, MIL-C-46168, MS-90495, 3431-95-CL-A04, and ASTM D3951-82.

b. The following reference documents have been added to this revision: BB-A-106, BB-C-101, BB-O-925, TT-P-664, PPP-B-1055, PPP-P-40, PPP-T-60, MIL-P-116, MIL-B-121, MIL-T-152, MIL-B-208, MIL-C-3600, MIL-C-5501, MIL-M-7866, MIL-C-53072, MIL-E-10062, MIL-A-18455, MIL-STD-129, MIL-STD-209, MIL-STD-171, MIL-STD-454, MIL-STD-810, MIL-STD-889, MIL-STD-1186, MIL-STD-1190, 11022380, 11022381, 13226E7140, 11021547, National Motor Freight Classification, Uniform Freight Classification, and AWS A3.0.

c. The material paragraph has been revised in this revision.

d. The weight of the shop has been changed in 3.4.1.1.

e. A lifting and tie-down requirement has been added per paragraph 3.4.2.1.

f. The rough handling paragraph has been changed to rail impact per paragraph 3.4.2.3.

g. The rough handling test has been changed to rail impact per paragraph 4.6.1.

h. A lifting and tie-down test has been added per paragraph 4.6.3.1.

i. The packaging requirements have been updated in Section 5 of this revision.

j. A subject term (key word) listing paragraph has been added to this revision.

k. This revision complies with current format of MIL-STD-961C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314



OFFICIAL BUSINESS

COMMANDER
US ARMY ARMAMENT RESEARCH DEVELOPMENT
AND ENGINEERING CENTER
ATTN: SMCAR-EST-S
ROCK ISLAND, IL 61299-7300

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:35Z
Check the source to verify that this is the current version before use.